

HAT-P-36b

R-1275

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_220204 · created 2026-07-31T14:20:03+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459614.9222 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1572 +/- 0.0049
Transit depth (Rp/Rs)^2	2.47 +/- 0.15 [%]
Orbital Inclination (inc)	89.69 +/- 1.17
Airmass coefficient 1 (a1)	1.21 +/- 0.014
Airmass coefficient 2 (a2)	-0.168 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	None
Optimal Aperture	4.02
Optimal Annulus	13.71
Transit Duration (day)	0.10042 +/- 0.00065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1572 $\pm$ 0.0049 vs 0.126 (deviation 4.94 σ)
Duration fit vs published	0.10042 d vs 0.0925 d (deviation 9.45 σ)
Mid-time O-C	7.4 min
Depth SNR	24.9
Residual scatter	1.18 %

Light curve

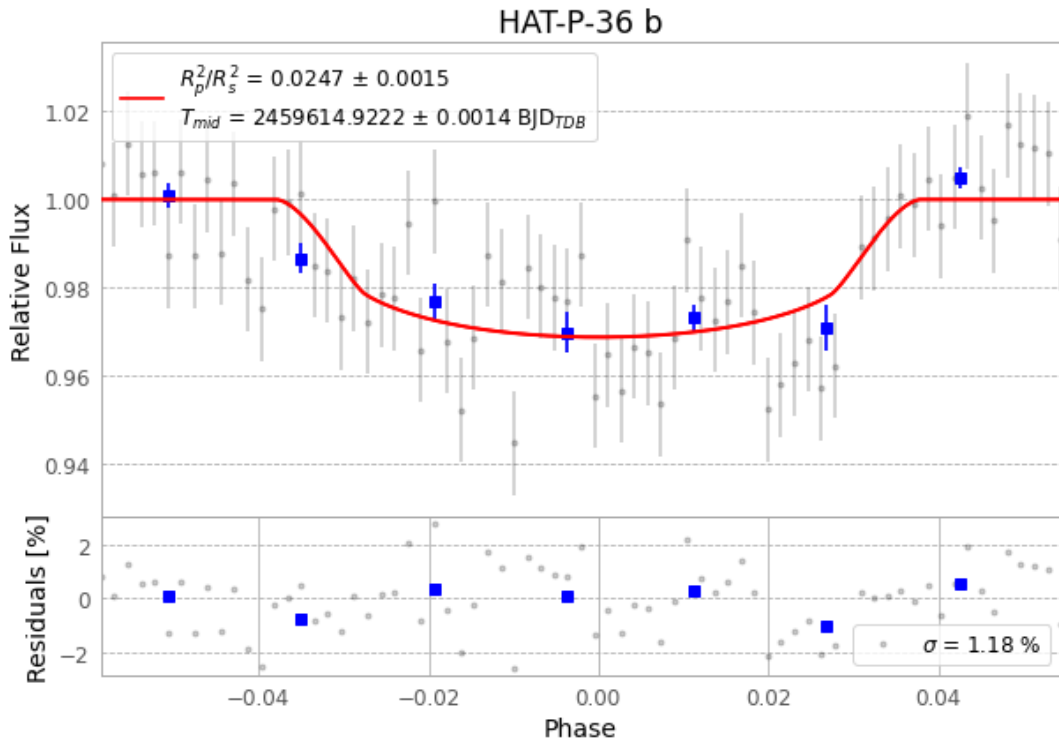


Figure 1 — FinalLightCurve_HAT-P-36 b_2022-02-04.png (SHA-256 0d3cab8947720759...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [230, 251], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 87f60d7a0f672437...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), HATP-36220204080912.FITS → HATP-36220204114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-220204.log (4ff183391198...), FinalLightCurve_HAT-P-36 b_2022-02-04.png (0d3cab894772...), FinalLightCurve_HAT-P-36 b_2022-02-04.csv (c241c8e9786a...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:20Z.